LESSON 8: ETHERNET FUNDAMENTALS

TRAINING FOCUS

This lesson is part of the FEMTECT Network Engineering Foundation course. It is written to help students build strong networking fundamentals through clear explanations, real enterprise context, practical troubleshooting logic, and certification-aware technical depth.

VISUAL DIAGRAM: ETHERNET FRAME DELIVERY

Visual flow

From	Relationship	To
PC1; Source MAC	connects to	Frame
Frame	connects to	Switch; Learns source MAC; Forwards by destination MAC
Switch; Learns source MAC; Forwards by destination MAC	connects to	PC2; Destination MAC

This visual separates the Ethernet frame from IP routing and shows why switches care about MAC addresses.

DEFINITION AND TECHNICAL IN-DEPTH KNOWLEDGE

8.1 Ethernet

Ethernet is the most widely used wired LAN technology in modern networks. It defines how devices format, address, transmit, and receive frames on a local network. Ethernet operates mainly at Layer 2, the Data Link layer, and depends on Layer 1 physical media such as copper cables, fiber optic cables, transceivers, and switch ports.

In Lesson 7, cabling and physical media were explained as the pathways that carry signals. Ethernet builds on those pathways by defining the frame format, MAC addressing, local delivery behavior, speed negotiation, duplex behavior, and switching foundation used in LANs.

Ethernet exists because devices on a local network need a standard way to communicate. A laptop, printer, server, IP phone, switch, and router interface must agree on how frames are structured and how local addresses are used. Ethernet provides that standard.

In real enterprise networks, Ethernet is used to connect end devices to access switches, switches to distribution switches, servers to data center switches, access points to wired networks, routers to LAN segments, and firewalls to internal and external networks. Even when users connect wirelessly, wireless access points often connect back to the wired network using Ethernet.

For CCNA and CompTIA Network+, students must understand Ethernet frame structure, MAC addressing, collision domains, broadcast domains, duplex, speed, and auto-negotiation. These concepts prepare students for enterprise switching, VLANs, trunking, Spanning Tree Protocol, and troubleshooting.

8.2 Technical and Business Importance of Ethernet

Ethernet matters because most LAN communication begins with an Ethernet frame. When a device sends traffic on a wired LAN, the data is encapsulated into an Ethernet frame. That frame contains local delivery information, including source and destination MAC addresses. Switches use this information to move traffic inside the LAN.

If Ethernet behavior fails, the network may appear broken even when IP addressing is correct. A PC may have a valid IP address but fail to communicate because the switch port is in the wrong VLAN. A server may have the correct gateway but experience poor performance because of duplex mismatch or physical errors. A broadcast

storm may overload a LAN. A wrong MAC address table entry may send traffic in the wrong direction. A damaged cable may cause frame errors.

Ethernet is also important because it introduces two major ideas that will appear repeatedly in the course: local delivery and segmentation. MAC addresses handle local delivery. Collision domains and broadcast domains help explain how network traffic is contained, separated, and controlled. VLANs, which will be covered later, are built on broadcast-domain control.

8.3 ETHERNET STANDARDS AND LAN OPERATION

Ethernet is standardized by IEEE 802.3. The standard has evolved over time from older shared-media networks to modern switched full-duplex networks. Early Ethernet networks used hubs and shared media, where devices competed to transmit. Modern Ethernet networks use switches, where each device typically has a dedicated switch port.

Ethernet can run over different physical media and speeds. Common examples include Fast Ethernet at 100 Mbps, Gigabit Ethernet at 1 Gbps, 10 Gigabit Ethernet, 40 Gigabit Ethernet, 100 Gigabit Ethernet, and faster standards in data centers and service provider networks.

Ethernet does not replace IP. Ethernet handles local frame delivery inside a LAN or VLAN. IP handles logical addressing and routing between networks. In a typical packet journey, an IP packet is carried inside an Ethernet frame on each local Ethernet segment.

Ethernet Characteristics

- Common wired LAN technology.
- Standardized by IEEE 802.3.
- Uses frames at Layer 2.
- Uses MAC addresses for local delivery.
- Commonly uses switches in modern LANs.
- Supports many speeds, including 100 Mbps, 1 Gbps, 10 Gbps, and higher.
- Works over copper and fiber media.

Enterprise Example

In an office, a user's PC connects to an access switch using Ethernet over CAT6 cable. The PC sends Ethernet frames to the switch. If the destination is local, the switch forwards the frame toward the destination device. If the destination is remote, the PC sends the frame to the default gateway's MAC address, and the router or Layer 3 switch routes the packet onward.

8.4 ETHERNET FRAME STRUCTURE

An Ethernet frame is the Layer 2 data unit used by Ethernet. It carries network-layer data, such as an IP packet, across a local Ethernet link. The frame includes addressing and control information that switches and network interfaces use to deliver and verify the data.

Ethernet frame structure matters because switches forward frames based on MAC addresses. If students understand the frame, they can understand why switches learn MAC addresses, why VLAN tags exist, why frame errors matter, and why Layer 2 troubleshooting is different from Layer 3 troubleshooting.

Main Ethernet Frame Fields

The common Ethernet frame includes these important fields:

- Preamble and Start Frame Delimiter.
- Destination MAC Address.

- Source MAC Address.
- EtherType or Length.
- Payload/Data.
- Frame Check Sequence.

Preamble and Start Frame Delimiter

The preamble and Start Frame Delimiter help receiving devices synchronize and identify the beginning of a frame. This is part of how Ethernet prepares the receiver to read the incoming bits correctly.

In normal troubleshooting, technicians rarely inspect the preamble directly, but it is important conceptually because it shows that Ethernet frames are carefully structured, not random signals.

Destination MAC Address

The destination MAC address identifies the local Layer 2 recipient of the frame. If the destination is on the same local network, the destination MAC may belong to the final receiving device. If the destination IP is remote, the destination MAC is usually the MAC address of the default gateway.

Switches examine the destination MAC address to decide where to forward the frame.

Source MAC Address

The source MAC address identifies the network interface that sent the frame on the local link. Switches learn MAC addresses by reading the source MAC address of incoming frames. This learning process allows switches to build MAC address tables. Lesson 9 will explain MAC address tables in more depth.

EtherType or Length

The EtherType field identifies the type of payload carried inside the frame. For example, EtherType `0x0800` indicates IPv4, and `0x86DD` indicates IPv6. This tells the receiving device what protocol should process the payload after the Ethernet header is removed.

Payload/Data

The payload is the data being carried by the Ethernet frame. In many modern networks, this payload is an IP packet. The payload may contain TCP, UDP, ICMP, or other upper-layer information inside the IP packet.

Frame Check Sequence

The Frame Check Sequence, or FCS, is used for error detection. It allows the receiving device to check whether the frame was corrupted during transmission. If the frame fails the FCS check, it is discarded.

FCS is important because physical media problems, electrical noise, bad cables, faulty transceivers, or duplex issues can cause corrupted frames. Switch interface counters may show CRC or FCS errors when Layer 1 or Layer 2 problems exist.

Ethernet Frame Structure Table

Field	Purpose
Preamble / SFD	Synchronizes receiver and marks frame start
Destination MAC	Identifies local Layer 2 destination
Source MAC	Identifies local Layer 2 sender
EtherType/Length	Identifies payload type or length
Payload/Data	Carries upper-layer data, commonly an IP packet
FCS	Detects frame corruption

Important Exam Note

For CCNA and Network+, remember that Ethernet frames use MAC addresses, while IP packets use IP addresses. Switches forward frames. Routers forward packets.

8.5 MAC ADDRESSING

A MAC address is a Layer 2 hardware address used to identify a network interface on a local network. MAC stands for Media Access Control. Ethernet devices use MAC addresses for local frame delivery.

A MAC address is 48 bits long and is commonly written as 12 hexadecimal digits. It may appear in formats such as:

Command / configuration block

```
00:1A:2B:3C:4D:5E
00-1A-2B-3C-4D-5E
001A.2B3C.4D5E
```

The first 24 bits are often associated with the vendor or manufacturer and are called the Organizationally Unique Identifier, or OUI. The remaining bits identify the specific interface, although modern devices and virtualization can sometimes use locally administered or randomized MAC addresses.

Why MAC Addresses Exist

MAC addresses exist because devices on the same local network need local identifiers. IP addresses identify logical network location, but Ethernet needs a Layer 2 address to deliver frames across the local segment.

For example, if a PC wants to send traffic to its default gateway, it places the gateway's MAC address in the destination MAC field of the Ethernet frame. The switch forwards the frame based on that MAC address. The IP packet inside the frame may be destined for a remote server, but the Ethernet frame is only concerned with the next local Layer 2 destination.

Types of MAC Addresses

Unicast MAC Address

A unicast MAC address identifies one specific network interface. Most normal Ethernet traffic uses unicast MAC addresses.

Broadcast MAC Address

The broadcast MAC address is:

Command / configuration block

```
FF:FF:FF:FF:FF:FF
```

It means all devices in the local broadcast domain should receive the frame. ARP Requests use broadcast at Layer 2.

Multicast MAC Address

A multicast MAC address identifies a group of receivers. It is used when frames should be delivered to selected devices that are interested in a multicast group.

MAC Address Troubleshooting

MAC addresses are useful in troubleshooting because they help engineers identify where a device is connected. On a switch, the MAC address table can show which port learned a device's MAC address. This helps locate devices, investigate loops, troubleshoot wrong VLANs, and detect unexpected devices.

Windows command:

Command / configuration block

```
getmac
ipconfig /all
```

Linux command:

Command / configuration block

```
ip link
```

Cisco IOS command:

Command / configuration block

```
show mac address-table
```

8.6 LOCAL DELIVERY: MAC ADDRESS VS IP ADDRESS

MAC addresses and IP addresses work together, but they do different jobs.

An IP address identifies the logical source and destination across networks. A MAC address identifies the local Layer 2 sender and local Layer 2 next destination. In a remote communication, the destination IP address may be the final server, but the destination MAC address is usually the default gateway's MAC address.

This distinction is one of the most important Ethernet concepts.

Example: Same LAN

PC1 at `192.168.10.10` sends traffic to PC2 at `192.168.10.20` in the same VLAN. PC1 uses ARP to learn PC2's MAC address. The Ethernet frame destination MAC is PC2's MAC address.

Example: Remote Network

PC1 at `192.168.10.10` sends traffic to a server at `10.20.5.50`. The server is not in the local network. PC1 uses ARP to learn the default gateway's MAC address. The Ethernet frame destination MAC is the gateway's MAC address, while the IP packet destination remains `10.20.5.50`.

MAC vs IP Comparison

Feature	MAC Address	IP Address
OSI Layer	Layer 2	Layer 3
Main Purpose	Local delivery	Logical addressing and routing
Used By	Switches and NICs	Routers, hosts, Layer 3 devices
Changes Across Routed Hops	Yes	Usually no, unless NAT is used
Example	`00:1A:2B:3C:4D:5E`	`192.168.10.10`

8.7 COLLISION DOMAINS

A collision domain is a network segment where devices share the same transmission space and where collisions can occur if two devices transmit at the same time. Collisions were common in older Ethernet networks that used hubs or shared media.

A collision happens when two devices transmit on a shared medium at the same time and the signals interfere with each other. Older Ethernet used CSMA/CD, which means Carrier Sense Multiple Access with Collision Detection. Devices listened before transmitting, detected collisions, waited a random time, and tried again.

Modern switched Ethernet greatly reduces collision problems. Each switch port is its own collision domain. In full-duplex switched Ethernet, collisions should not occur because the device and switch can send and receive simultaneously over separate signaling paths.

Hub vs Switch Collision Behavior

A hub is a Layer 1 device that repeats signals to all ports. All devices connected to a hub share one collision domain. If two devices transmit at the same time, a collision can occur.

A switch is a Layer 2 device that makes forwarding decisions. Each switch port is a separate collision domain. This is one reason switches replaced hubs in modern LANs.

Collision Domain Examples

Device/Design	Collision Domain Behavior
Hub with 10 connected PCs	One shared collision domain
Switch with 10 connected PCs	10 separate collision domains
Full-duplex switch port	Collisions should not occur

Collision Domain Troubleshooting Clues

- Collisions are mostly historical in modern full-duplex switched networks.
- Late collisions usually indicate duplex mismatch or physical-layer issues.
- If collisions appear on a modern switch port, check duplex settings and cabling.
- Hubs should not be used in modern enterprise LANs.

8.8 BROADCAST DOMAINS

A broadcast domain is a network area where broadcast frames are forwarded. If one device sends a Layer 2 broadcast, all devices in the same broadcast domain receive it.

Broadcast domains matter because broadcast traffic consumes network resources. Broadcasts are useful for discovery protocols such as ARP and DHCP, but excessive broadcasts can reduce performance. Large flat networks with too many devices in one broadcast domain are harder to manage and less secure.

Switches forward broadcasts out all ports in the same VLAN, except the port where the broadcast was received. Routers do not forward Layer 2 broadcasts by default. VLANs create separate broadcast domains inside a switched network.

Broadcast Domain Examples

Design	Broadcast Domain Behavior
One switch, one VLAN	One broadcast domain
One switch, three VLANs	Three broadcast domains
Two switches connected by trunk carrying one VLAN	One broadcast domain for that VLAN across both switches
Router separating two LANs	Two broadcast domains

Broadcast Domain Troubleshooting Clues

- Excessive broadcast traffic can slow a network.
- ARP and DHCP use broadcast behavior in IPv4 networks.
- VLANs control broadcast scope.
- If DHCP does not cross a router, DHCP relay/IP helper may be required.
- If a device is in the wrong VLAN, it may be in the wrong broadcast domain.

Enterprise Example

A company places staff, voice, CCTV, and guest Wi-Fi in separate VLANs. Each VLAN is a separate broadcast domain. Guest users' broadcasts do not reach staff computers, and CCTV device broadcasts do not flood the voice VLAN. This improves performance, segmentation, and troubleshooting.

8.9 COLLISION DOMAINS VS BROADCAST DOMAINS

Collision domains and broadcast domains are different concepts.

A collision domain is about where collisions can occur on shared media. It is mostly related to older Ethernet behavior and hub-based networks. A broadcast domain is about where broadcast frames are delivered. Broadcast domains are still very important in modern switched networks.

Collision vs Broadcast Domain Comparison

Feature	Collision Domain	Broadcast Domain
Main Concern	Simultaneous transmissions causing collisions	Broadcast traffic reach
Commonly Controlled By	Switch ports and full-duplex links	Routers and VLANs
Still Important Today	Less common, mostly for troubleshooting duplex issues	Very important
Hub Behavior	One shared collision domain	One broadcast domain unless routed
Switch Behavior	Each port is separate collision domain	VLAN defines broadcast domain
Router Behavior	Separates collision domains by interface	Separates broadcast domains by interface

Exam Reminder

For CCNA and Network+, remember: each switch port is a collision domain, and each VLAN is a broadcast domain. Routers separate broadcast domains.

8.10 DUPLEX

Duplex describes whether a network interface can send and receive at the same time.

Half-duplex means a device cannot send and receive simultaneously. It must take turns. Half-duplex was common with older hub-based Ethernet. Collisions can occur in half-duplex environments.

Full-duplex means a device can send and receive at the same time. Modern switched Ethernet normally uses full-duplex. In full-duplex communication, collisions should not occur.

Duplex matters because mismatched duplex settings can cause serious performance problems. A duplex mismatch may occur when one side is forced to full-duplex and the other side auto-negotiates or uses half-duplex. The link may come up, but performance may be poor, with errors, retransmissions, and late collisions.

Duplex Comparison

Duplex Mode	Behavior	Collision Risk	Modern Use
Half-duplex	Send or receive, not both at the same time	Yes	Mostly legacy
Full-duplex	Send and receive simultaneously	No expected collisions	Standard modern Ethernet

Duplex Troubleshooting Clues

- Slow performance with link up may indicate duplex mismatch.
- Late collisions suggest duplex or physical-layer problems.
- CRC errors may indicate cabling, physical media, or duplex issues.

- Modern devices should usually use auto-negotiation unless design requires otherwise.

8.11 SPEED AND AUTO-NEGOTIATION

Ethernet speed describes the data rate supported by the link. Common Ethernet speeds include 10 Mbps, 100 Mbps, 1 Gbps, 10 Gbps, 40 Gbps, 100 Gbps, and higher in modern data centers.

Auto-negotiation is the process where two connected Ethernet devices exchange capabilities and agree on speed and duplex. For example, a PC and switch may negotiate 1 Gbps full-duplex if both support it and the cable is suitable.

Auto-negotiation exists because devices may support different speeds and duplex modes. It allows them to choose the best common setting automatically. In modern networks, auto-negotiation usually works well when cabling and hardware are correct.

However, problems can occur. A bad cable may cause a device to negotiate at 100 Mbps instead of 1 Gbps. A manually forced speed or duplex on one side can create mismatch. An older device may not support the desired speed. A transceiver may not match the port speed.

Common Speed and Negotiation Symptoms

- Link comes up at 100 Mbps instead of 1 Gbps.
- Link flaps between up and down.
- Interface shows high errors.
- One side shows full-duplex and the other shows half-duplex.
- Fiber link does not come up due to optic mismatch.

Cisco IOS Speed/Duplex Checks

Command / configuration block

```
show interfaces status
show interfaces
```

Windows Checks

Command / configuration block

```
Get-NetAdapter
```

Linux Checks

Command / configuration block

```
ethtool <interface>
```

Best Practice

Use auto-negotiation in most modern Ethernet environments unless there is a specific engineering requirement to hard-code speed and duplex. If hard-coding is required, configure both sides consistently and document the reason.

8.12 ETHERNET TROUBLESHOOTING

Ethernet troubleshooting combines Layer 1 and Layer 2 thinking. The technician must check the physical link, negotiated speed, duplex, MAC learning, VLAN placement, broadcast behavior, and interface errors.

Common Ethernet Troubleshooting Flow

1. Check link light and cable.

2. Check interface status.

3. Check speed and duplex.

4. Check error counters such as CRC, input errors, output errors, and late collisions.

5. Check MAC address learning.

6. Check VLAN assignment.

7. Check whether broadcasts are excessive.

8. Check whether the issue affects one device, one switch port, one VLAN, or many users.

Cisco IOS Ethernet Troubleshooting Commands

Command / configuration block

```
show interfaces status
show interfaces
show mac address-table
show vlan brief
show interfaces counters errors
```

Windows and Linux Checks

Command / configuration block

```
Get-NetAdapter
ipconfig /all
```

Command / configuration block

```
ip link
ethtool <interface>
```

Important Troubleshooting Reminder

If a link is up but performance is poor, do not assume the network is healthy. Check speed, duplex, errors, cable quality, and interface counters.

8.13 CYBERSECURITY RELEVANCE

Ethernet security matters because Layer 2 is the first point of access in many wired networks. If an attacker can plug into an Ethernet port, they may attempt to join the LAN, capture traffic, spoof MAC addresses, send rogue DHCP messages, perform ARP spoofing, or attack local services.

MAC addresses are not strong identity by themselves. They can be spoofed. This means security should not depend only on trusting a MAC address. Enterprise networks use additional controls such as port security, 802.1X authentication, DHCP Snooping, Dynamic ARP Inspection, VLAN segmentation, disabled unused ports, and monitoring.

Broadcast domains also matter for security. A flat network with one large broadcast domain allows more devices to see discovery traffic and may allow attacks to spread more easily. Segmentation using VLANs reduces unnecessary exposure.

Security Reminder

Ethernet is local, but local does not mean safe. Physical access plus weak Layer 2 controls can become a serious security risk.

8.14 CLOUD NETWORKING RELEVANCE

Cloud customers usually do not manage physical Ethernet switches inside the cloud provider's data center, but Ethernet concepts still matter. Cloud providers use high-speed Ethernet fabrics inside their infrastructure. Virtual NICs, virtual switches, overlays, and software-defined networks still depend on Layer 2 and Layer 3 principles.

In cloud environments, customers often see virtual MAC addresses, virtual NICs, virtual subnets, and security policies instead of physical switch ports. The customer may not troubleshoot a physical Ethernet cable, but they still troubleshoot interface attachment, IP assignment, security groups, virtual routing, and traffic flow.

Hybrid networks also depend on Ethernet locally. A company may access cloud services through an office LAN, switch, firewall, ISP handoff, and router. If the local Ethernet link to the firewall is negotiating incorrectly or dropping frames, cloud applications may appear slow even though the cloud platform is healthy.

Cloud Reminder

Cloud abstracts many Layer 2 details, but Ethernet concepts remain important for hybrid networks, virtualization, data centers, and troubleshooting local access to cloud services.

9. KEY TERMS / GLOSSARY

Term	Definition
Ethernet	Common wired LAN technology standardized by IEEE 802.3.
Ethernet Frame	Layer 2 data unit used by Ethernet.
MAC Address	48-bit Layer 2 address used for local Ethernet delivery.
OUI	Organizationally Unique Identifier, vendor-related part of a MAC address.
Preamble	Frame field used to help synchronize the receiver.
Start Frame Delimiter	Field marking the start of an Ethernet frame.
EtherType	Field identifying the protocol carried in the Ethernet payload.
Payload	Data carried inside an Ethernet frame, commonly an IP packet.
FCS	Frame Check Sequence used for error detection.
Collision Domain	Network area where collisions can occur.
Broadcast Domain	Network area where broadcast frames are forwarded.
Hub	Layer 1 device that repeats signals to all ports.
Switch	Layer 2 device that forwards frames using MAC addresses.
CSMA/CD	Older Ethernet collision detection method.
Half-Duplex	Communication mode where a device cannot send and receive at the same time.
Full-Duplex	Communication mode where a device can send and receive simultaneously.
Auto-Negotiation	Process where devices agree on speed and duplex.
CRC Error	Error indicating possible frame corruption, often related to physical issues.
Late Collision	Collision detected late, often related to duplex mismatch or legacy problems.
Broadcast MAC	`FF:FF:FF:FF:FF:FF`, used to send to all local devices in a broadcast domain.

10. COMPARISON TABLES

10.1 Ethernet Frame Fields

Field	Main Purpose
Preamble / SFD	Synchronization and frame start
Destination MAC	Local Layer 2 destination
Source MAC	Local Layer 2 sender
EtherType/Length	Identifies payload type or length
Payload	Carries upper-layer data
FCS	Detects frame corruption

10.2 MAC Address vs IP Address

Feature	MAC Address	IP Address
Layer	Layer 2	Layer 3
Used For	Local frame delivery	Logical addressing and routing
Device Mainly Using It	Switch	Router
Changes Across Routed Hops	Yes	Usually no, unless NAT
Example Format	`001A.2B3C.4D5E`	`192.168.10.10`

10.3 Collision Domain vs Broadcast Domain

Feature	Collision Domain	Broadcast Domain
Concern	Collisions on shared media	Broadcast traffic reach
Controlled By	Switch ports/full duplex	VLANs and routers
Modern Importance	Lower, mostly troubleshooting	High
Hub Effect	One shared collision domain	One broadcast domain
Switch Effect	Each port is separate collision domain	VLAN defines broadcast domain

10.4 Half-Duplex vs Full-Duplex

Feature	Half-Duplex	Full-Duplex
Send and Receive	One direction at a time	Both directions at once
Collisions	Possible	Not expected
Common Today	Rare/legacy	Standard
Common Problem	Collisions	Mismatch if configured badly

10.5 Common Ethernet Symptoms

Symptom	Possible Cause
No link	Cable, port, NIC, transceiver, shutdown port
Link at wrong speed	Bad cable, old device, negotiation issue
High CRC errors	Cable, interference, transceiver, physical problem
Late collisions	Duplex mismatch or legacy shared media
Broadcast storm	Loop or excessive broadcast traffic
Device unreachable in same VLAN	MAC learning, VLAN, cable, NIC, local firewall

11. REAL-WORLD EXAMPLE / ANALOGY

Think of Ethernet like an office mailroom inside one building.

The IP address is like the full company address used to reach a building or branch. The MAC address is like the desk or room label used inside the local office. The Ethernet frame is like the internal envelope used by the mailroom to deliver the package locally. If the package must go to another branch, the local mailroom first sends it to the outgoing mail desk, which is like the default gateway.

Collision domains are like shared walkways where people can bump into each other if everyone moves at once. Modern switches create separate paths, reducing collisions. Broadcast domains are like announcement zones. If an announcement is made in one department, everyone in that department hears it. VLANs create separate announcement zones so every message does not disturb the whole building.

13. WINDOWS, LINUX, AND CISCO IOS PRACTICAL COMMANDS

Windows Commands

Command / configuration block

```
getmac
ipconfig /all
Get-NetAdapter
arp -a
```

Linux Commands

Command / configuration block

```
ip link
ip addr
ip neigh
ethtool <interface>
```

Cisco IOS Commands

Command / configuration block

```
show mac address-table
show interfaces status
show interfaces
show vlan brief
show interfaces counters errors
```

14. QUICK-RECAP CHEAT SHEET

- Ethernet is the most common wired LAN technology.
- Ethernet is standardized by IEEE 802.3.
- Ethernet uses frames at Layer 2.
- Ethernet uses MAC addresses for local delivery.
- A MAC address is 48 bits long.
- The broadcast MAC address is `FF:FF:FF:FF:FF:FF`.
- Switches forward frames based on MAC addresses.
- Routers forward packets based on IP addresses.
- An Ethernet frame includes destination MAC, source MAC, EtherType, payload, and FCS.
- FCS helps detect frame corruption.
- A collision domain is where collisions can occur.

- Each switch port is its own collision domain.
- A broadcast domain is where broadcast frames are forwarded.
- Each VLAN is a broadcast domain.
- Routers separate broadcast domains.
- Half-duplex cannot send and receive at the same time.
- Full-duplex can send and receive at the same time.
- Modern switched Ethernet normally uses full-duplex.
- Auto-negotiation allows devices to agree on speed and duplex.
- Duplex mismatch can cause poor performance and errors.

7. LESSON ASSESSMENT

This assessment is intentionally focused. Students should answer clearly, use correct networking terminology, and show practical troubleshooting thinking.

7.1 Theory Questions

1. Explain the main networking concept covered in this lesson and describe its role in a real enterprise network.

2. Describe two technical problems that can occur when this lesson's concept is poorly designed or incorrectly configured.

3. Explain how a network technician or junior network engineer would verify that this part of the network is working correctly.

4. Describe one security, reliability, or performance concern related to this topic.

5. Explain how this topic connects to CCNA, Network+, IT support, or junior network engineering work.

7.2 Multiple-Choice Questions

1. What is the best reason for studying this lesson's topic?

A. It helps students understand and support real network operation. B. It replaces the need for troubleshooting. C. It removes the need for IP addressing. D. It is only useful for memorizing definitions.

2. When troubleshooting a problem related to this lesson, what should be done first?

A. Gather evidence and identify the exact scope of the problem. B. Replace every network device immediately. C. Delete the running configuration without backup. D. Assume the Internet service provider is always responsible.

3. Which habit best reflects professional network engineering practice?

A. Verify changes with commands, tests, and documented results. B. Make random changes until something works. C. Ignore diagrams and rely only on memory. D. Avoid documenting successful fixes.

4. Which item is most useful when proving that a network feature is working?

A. A successful test result compared with the expected result. B. A guess from another user. C. A device name with no configuration evidence. D. A screenshot unrelated to the problem.

5. What should students connect this lesson to during hands-on practice?

A. Addressing, device roles, commands, verification, and troubleshooting. B. Only typing speed. C. Only cable color. D. Only memorizing vendor logos.

6. In an enterprise environment, why is documentation important for this topic?

A. It helps future troubleshooting, audits, handover, and change control. B. It makes testing unnecessary. C. It disables security risks automatically. D. It replaces the need for correct configuration.

7. Which approach gives the strongest learning outcome?

A. Combining technical explanation with lab practice and verification. B. Reading commands without understanding them. C. Skipping troubleshooting steps. D. Memorizing answers without using the network.

8. If a configuration change causes a new outage, what should the technician use?

A. A rollback plan and saved baseline information. B. A random new subnet. C. A different monitor. D. A guess without evidence.

9. Which result shows that a student understands the lesson at a practical level?

A. The student can explain the concept, configure or identify it, test it, and troubleshoot failure. B. The student can only repeat the title. C. The student avoids using commands. D. The student ignores enterprise examples.

10. Why are scenario questions included in this course?

A. They train students to apply concepts to real network problems. B. They remove the need for labs. C. They focus only on spelling. D. They are unrelated to IT jobs.

7.3 Scenario-Based Questions

1. A user or department reports a connectivity problem related to this lesson's topic. Write the first five checks you would perform, the commands or tools you would use, and the evidence you expect to collect. 2. You are asked to explain this topic to a new IT support intern before a hands-on lab. Write a short technical explanation that connects the concept to real devices, real traffic flow, and troubleshooting.

7.4 Answer Guide

Theory answers should be marked for technical accuracy, clear explanation, correct terminology, and practical relevance to enterprise networking. Scenario answers should show a logical troubleshooting sequence, evidence gathering, likely causes, verification steps, and a professional final recommendation.

Multiple-choice answers: 1. A; 2. A; 3. A; 4. A; 5. A; 6. A; 7. A; 8. A; 9. A; 10. A